The Phase 1 Winner-Take-All Balance Term

Soft territory, the balance penalty and its gradient, the discrete-area trim, and Γ-consistency

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Abstract

This document derives the *territory-aware* additions to the Phase 1 relaxation energy of `ProjectedGradientOptimizer` (`src/optimization/pgd_optimizer.py`): a differentiable *soft winner-take-all territory* T_k per cell built from power-normalized weights, the *balance penalty* $P_{\text{bal}} = \frac{\gamma}{2}\sum_k r_k^2$ with $r_k = (T_k - \bar{A})/\bar{A}$, its exact analytical gradient, and the *discrete-area trim* — a damped controller on the projection's per-cell area targets whose fixed point is exact discrete (argmax) area equality. We prove the six structural properties the design relies on (total-area consistency, self-deactivation, interface-band localization, neutrality at the symmetric state, restoration of starving cells, and Γ-consistency), justify the default sharpening exponent $p = 2$, and record the one-time calibration of the strength γ. All gradients are analytical and verified against central finite differences to relative error $< 10^{-6}$ (`testing/test_wta_balance_gradient_analytical.py`).

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1 Overview

Phase 1 (see `docs/math/06-phase1-energy-discretization/` for the base energy) enforces equal *continuous* cell masses $\sum_i v_i u_{i,k} = \bar{A}$, but the deliverable of the relaxation is the *winner-take-all* (WTA) partition: vertex i belongs to cell $\arg\max_k u_{i,k}$. Nothing in the base energy or the constraints references which side of the argmax boundary a unit of interface-band mass sits on, so at high cell counts N the discrete (argmax) territories drift from equal even though the continuous masses are pinned. The measured diagnosis and the design rationale live in `docs/plans/PHASE1_N1000_VALIDITY_PLAN.md`; the root cause is the following exact accounting identity.

Proposition 1.1 (Accounting identity)**.** *Let $U \in \mathbb{R}^{|V|\times N}$ be feasible (rows on the simplex, columns mass-pinned: $\sum_k u_{i,k} = 1$, $u_{i,k} \geq 0$, $\sum_i v_i u_{i,k} = \bar{A}$). Let $\omega(i) = \arg\max_k u_{i,k}$, and define per cell k the WTA territory, the lost mass, and the gained mass*

$$T_k^{\mathrm{wta}} = \sum_{i:\,\omega(i)=k} v_i, \qquad \mathrm{lost}_k = \sum_{i:\,\omega(i)\neq k} v_i\, u_{i,k}, \qquad \mathrm{gain}_k = \sum_{i:\,\omega(i)=k} v_i\big(1-u_{i,k}\big).$$

Then

$$T_k^{\mathrm{wta}} - \bar{A} \;=\; \mathrm{gain}_k - \mathrm{lost}_k. \tag{1}$$

Proof. Using $\sum_l u_{i,l} = 1$ on each winning vertex, $T_k^{\mathrm{wta}} = \sum_{i:\,\omega(i)=k} v_i \sum_l u_{i,l} = \sum_{i:\,\omega(i)=k} v_i u_{i,k} + \mathrm{gain}_k$. The first sum is the cell's own mass on won vertices, i.e. $\bar{A} - \mathrm{lost}_k$ by the mass constraint. □

The imbalance $T_k^{\mathrm{wta}} - \bar{A}$ is therefore exactly the cell's net interface-band mass exchange $\mathrm{gain}_k - \mathrm{lost}_k$ — a quantity the base energy prices *symmetrically* about $u = \frac{1}{2}$ and the projection never sees. The additions derived here couple it back into the optimality conditions:

1. a smooth penalty P_{bal} on a differentiable surrogate $T_k \approx T_k^{\mathrm{wta}}$ (Sections 3–6), added to the energy; and
2. a periodic, derivative-free retargeting of the projection's per-cell area targets d whose fixed point is *exact* discrete equality (Section 8).

Both are flag-gated and default-off; with the flags off the Phase 1 energy is byte-for-byte the one derived in `docs/math/06-phase1-energy-discretization/`.

2 Notation and setup

We reuse the Phase 1 notation of `docs/math/06-phase1-energy-discretization/`. The mesh has $|V|$ vertices with lumped-mass (area) weights $v_i > 0$ (`TriMesh.v`), total area $A = \sum_i v_i$, cell count N, and equal-area target $\bar{A} = A/N$. The density matrix is $U \in \mathbb{R}^{|V|\times N}$ with entries $u_{i,k} \in [0,1]$ (in the code: `phi = x.reshape(N, n)`, vertices as rows, cells as columns). The projection (`src/optimization/projection.py`) enforces

$$\sum_k u_{i,k} = 1 \;\; \forall i, \qquad 0 \leq u_{i,k} \leq 1, \qquad \sum_i v_i\, u_{i,k} = d_k \;\; \forall k, \tag{2}$$

where $d \in \mathbb{R}^N$ is the per-cell area-target vector, by default $d = \bar{A}\mathbf{1}$. The base energy (unchanged by this document) is

$$E_0(U) = \sum_k \Big[\varepsilon\, u_k^\top K u_k + \tfrac{1}{\varepsilon}\, q_k^\top M q_k\Big] + \lambda \sum_k \Big(1 - \tfrac{\mathrm{Var}(u_k)}{T_{\mathrm{eff}}}\Big), \qquad q_k = u_k \odot (1-u_k). \tag{3}$$

This document adds $E = E_0 + P_{\mathrm{bal}}$.

Remark 2.1 (Total area symbol)**.** Document 06 writes W for the total area; here we write A (and reserve W for nothing) to match the implementation plan. The row-simplex constraint in (2) makes each row $u_{i,\cdot}$ a point of the standard simplex Δ^{N-1}; all bounds below exploit only this.

3 Soft territory

The hard WTA territory $T_k^{\mathrm{wta}} = \sum_{i:\,\omega(i)=k} v_i$ is piecewise constant in U (its derivative is zero a.e. and undefined on the tie set), so it cannot enter a gradient method directly. We replace the hard argmax indicator by a *power-normalized (Gibbs-type) soft assignment.*

Definition 3.1 (Power-normalized weights and soft territory)**.** Fix a sharpening exponent $p \geq 1$ (default $p = 2$). For each vertex i and cell k,

$$w_{i,k} \;=\; \frac{u_{i,k}^p}{S_i}, \qquad S_i \;=\; \sum_{l=1}^{N} u_{i,l}^p, \qquad \text{so} \qquad \sum_k w_{i,k} = 1, \tag{4}$$

$$T_k \;=\; \sum_{i=1}^{|V|} v_i \, w_{i,k}. \tag{5}$$

Proposition 3.1 (Bounds on the normalizer)**.** *On the row simplex ($u_{i,\cdot} \in \Delta^{N-1}$), for every $p \geq 1$,*

$$N^{1-p} \;\leq\; S_i \;\leq\; 1. \tag{6}$$

Proof. Upper bound: $u_{i,l} \leq \sum_m u_{i,m} = 1$, so $u_{i,l}^p \leq u_{i,l}$ and $S_i \leq \sum_l u_{i,l} = 1$. Lower bound: $x \mapsto x^p$ is convex, so by Jensen $\frac{1}{N} S_i \geq \big(\frac{1}{N}\sum_l u_{i,l}\big)^p = N^{-p}$. Both are attained: the upper at one-hot rows, the lower at the uniform row. $\square$

S_i is therefore bounded away from zero *on the feasible set*, and $w_{i,\cdot}$ is a smooth (C^∞ in u for integer p, wherever $S_i > 0$) partition of unity. The implementation nevertheless guards $S_i \leftarrow \max(S_i, \mathrm{tiny})$ defensively, because the gradient may be evaluated on pre-projection iterates that violate (2).

Proposition 3.2 (Sharpening toward the argmax)**.** *For $p > 1$ and any row $u_{i,\cdot} \in \Delta^{N-1}$, the winner's share weakly increases and every non-winner's ratio to the winner strictly decreases:*

$$w_{i,\omega(i)} \geq u_{i,\omega(i)}, \qquad \frac{w_{i,l}}{w_{i,\omega(i)}} = \Big(\frac{u_{i,l}}{u_{i,\omega(i)}}\Big)^p \leq \frac{u_{i,l}}{u_{i,\omega(i)}},$$

with equality only at ties or one-hot rows. As $p \to \infty$, $w_{i,\cdot}$ converges to the uniform distribution on the argmax set (the hard indicator when the winner is unique); at $p = 1$, $w = u$ identically.

Proof. $S_i = \sum_l u_{i,l}\, u_{i,l}^{p-1} \leq u_{i,\omega(i)}^{p-1} \sum_l u_{i,l} = u_{i,\omega(i)}^{p-1}$, hence $w_{i,\omega(i)} = u_{i,\omega(i)}^p / S_i \geq u_{i,\omega(i)}$. The ratio identity is immediate from (4); for $p \to \infty$ divide numerator and denominator by $u_{i,\omega(i)}^p$. $\square$

Remark 3.1 (Why $p = 2$ is the default, and $p = 1$ is the blind spot)**.** At $p = 1$ the soft territory degenerates to the *continuous mass*, $T_k = \sum_i v_i u_{i,k} = d_k$, which the projection pins at $\bar{A}$: the term is identically zero on the feasible set. This is precisely the existing blind spot — the equal-mass constraint cannot see territory. $p = 2$ is the smallest integer exponent that (a) strictly sharpens toward the argmax (Proposition 3.2), (b) keeps $w_{i,\cdot}$ a smooth partition of unity with $S_i \geq 1/N$ (Proposition 3.1), and (c) yields the closed-form, bounded gradient of Section 5 (Proposition 5.3). Larger p sharpens more but steepens the gradient near crisp states and worsens conditioning; p is exposed as the config knob `wta_balance_power` (default 2.0).

Remark 3.2 (Alternative surrogate: temperature softmax)**.** The temperature softmax $w_{i,k} = e^{u_{i,k}/\tau}/\sum_l e^{u_{i,l}/\tau}$ is also a partition of unity (so Property 6.1 would hold), but it is not used, for two reasons. First, it never reaches a one-hot weight at a crisp vertex: at $u_{i,\cdot} = e_k$ it gives $w_{i,k} = e^{1/\tau}/(e^{1/\tau} + N - 1) < 1$, so the interior force of Property 6.3 vanishes only up to an $O\big(Ne^{-1/\tau}\big)$ residual, whereas power weights are *exactly* one-hot there. Second, its sharpening saturates at the fixed ratio $e^{1/\tau}$ on the bounded density range $[0, 1]$, so approaching the argmax requires $\tau \to 0$, which inflates the gradient's Lipschitz constant like $1/\tau$; the power weights sharpen exactly at crisp states with the bounded gradient of Proposition 5.3 and introduce no extra scale parameter.

4 The balance penalty

Definition 4.1 (Relative territory deviation and balance penalty)**.**

$$r_k \;=\; \frac{T_k - \bar{A}}{\bar{A}} \qquad \text{(dimensionless)}, \qquad P_{\text{bal}}(U) \;=\; \frac{\gamma}{2}\sum_{k=1}^{N} r_k^2, \qquad \gamma \geq 0. \tag{7}$$

The strength γ (`wta_balance_gamma`) is calibrated once in Section 7. The deviations are *relative* — everything downstream is in units of the fair share $\bar{A}$ — which is what makes one γ portable across N (Section 7).

5 The gradient

Only vertex i's own weights depend on $u_{i,k}$, so the derivative is local in i and couples all cells at that vertex through the normalizer S_i.

Proposition 5.1 (Weight and territory derivatives)**.** *For all cells l, k and vertices i,*

$$\frac{\partial w_{i,l}}{\partial u_{i,k}} \;=\; \frac{p\,u_{i,k}^{p-1}}{S_i}\,\big(\delta_{lk} - w_{i,l}\big), \tag{8}$$

$$\frac{\partial T_l}{\partial u_{i,k}} \;=\; v_i\,\frac{p\,u_{i,k}^{p-1}}{S_i}\,\big(\delta_{lk} - w_{i,l}\big). \tag{9}$$

Proof. From $w_{i,l} = u_{i,l}^p/S_i$ and $\partial S_i/\partial u_{i,k} = p\,u_{i,k}^{p-1}$, the quotient rule gives

$$\frac{\partial w_{i,l}}{\partial u_{i,k}} = \frac{p\,u_{i,k}^{p-1}\,\delta_{lk}\,S_i - u_{i,l}^p\,p\,u_{i,k}^{p-1}}{S_i^2} = \frac{p\,u_{i,k}^{p-1}}{S_i}\left(\delta_{lk} - \frac{u_{i,l}^p}{S_i}\right),$$

which is (8); (9) follows since T_l is linear in the weights with coefficients v_i and $w_{j,l}$ with $j \neq i$ does not depend on $u_{i,k}$. □

Proposition 5.2 (Gradient of the balance penalty)**.** *Define the per-vertex mean deviation $m_i = \sum_l r_l\,w_{i,l}$ (a convex combination of the r_l). Then*

$$\frac{\partial P_{\text{bal}}}{\partial u_{i,k}} \;=\; \frac{\gamma}{\bar{A}}\sum_l r_l\frac{\partial T_l}{\partial u_{i,k}} \;=\; \frac{\gamma\,p}{\bar{A}}\;v_i\;\frac{u_{i,k}^{p-1}}{S_i}\;\big(r_k - m_i\big). \tag{10}$$

For the default $p = 2$,

$$\frac{\partial P_{\text{bal}}}{\partial u_{i,k}} \;=\; \frac{2\gamma}{\bar{A}}\;v_i\;\frac{u_{i,k}}{S_i}\;\big(r_k - m_i\big). \tag{11}$$

Proof. Chain rule on (7): $\partial P_{\text{bal}}/\partial u_{i,k} = \gamma \sum_l r_l\, \partial r_l/\partial u_{i,k} = (\gamma/\bar{A}) \sum_l r_l\, \partial T_l/\partial u_{i,k}$. Insert (9) and split the δ_{lk} and $w_{i,l}$ parts: $\sum_l r_l(\delta_{lk} - w_{i,l}) = r_k - m_i$. □

Proposition 5.3 (Entry-wise gradient bound)**.** *On the feasible set, for $p = 2$,*

$$\left|\frac{\partial P_{\text{bal}}}{\partial u_{i,k}}\right| \le \frac{2\gamma\, v_i}{\bar{A}} \sqrt{N}\ \operatorname{osc}(r), \qquad \operatorname{osc}(r) = \max_l r_l - \min_l r_l. \tag{12}$$

Proof. Since m_i is a convex combination of the r_l, $|r_k - m_i| \le \operatorname{osc}(r)$. For the factor $u_{i,k}/S_i$: by Proposition 3.1 and $S_i \ge u_{i,k}^2$,

$$\frac{u_{i,k}}{S_i} \le \frac{u_{i,k}}{\max\!\big(u_{i,k}^2,\ 1/N\big)} = \min\!\Big(\frac{1}{u_{i,k}},\ N u_{i,k}\Big) \le \sqrt{N},$$

the last step at the crossover $u_{i,k} = N^{-1/2}$. □

The gradient is thus uniformly bounded on the feasible set — no barrier-like blow-up anywhere, including crisp states.

Vectorized assembly. With $U \in \mathbb{R}^{|V|\times N}$ the whole computation is two $O(|V|N)$ elementwise passes and two reductions (one over cells, one over vertices):

$$\begin{aligned}
&U^p \to S \to W = U^p/S \quad \text{(rows)}, \qquad T = (v\odot)\text{-weighted column sums of } W,\\
&r = (T - \bar{A})/\bar{A}, \qquad P_{\text{bal}} = \tfrac{\gamma}{2}\, r^\top r, \qquad m = Wr \quad \text{(per-vertex reduction)},\\
&\nabla_U P_{\text{bal}} = \frac{\gamma p}{\bar{A}}\, v \odot \frac{U^{p-1}}{S} \odot \big(\mathbf{1}\, r^\top - m\, \mathbf{1}^\top\big),
\end{aligned}$$

broadcasting v and S over columns and r, m over rows/columns respectively. This is the same cost class $O(|V|N)$ as the base energy.

Code: WTA balance term and gradient

`src/optimization/pgd_optimizer.py`: `compute_energy` / `compute_gradient`, guarded by `wta_balance_enabled` (default off; zero overhead when off)

```
Up = phi**p; S = np.maximum(Up.sum(1), tiny); W = Up / S[:, None]
T = W.T @ v; r = (T - Abar) / Abar; P_bal = 0.5 * gamma * (r @ r)
m = W @ r
G += (gamma * p / Abar) * v[:, None] * (phi**(p-1) / S[:, None]) * (r[None, :] -
m[:, None])
```

Config: `wta_balance_enabled`, `wta_balance_gamma` (γ), `wta_balance_power` (p) in `RelaxationConfig` (`src/pipeline/relaxation.py`). FD check: `testing/test_wta_balance_gradient_analytical.py`.

6 Structural properties

Proposition 6.1 (1. Total-area consistency)**.** $\displaystyle\sum_k T_k = \sum_i v_i \sum_k w_{i,k} = \sum_i v_i = A$ *exactly, for every feasible or infeasible U with $S_i > 0$. Consequently $\sum_k r_k = 0$: the deviation vector always lies in the zero-mean subspace, so P_{bal} measures only the* spread *of territories and is compatible with the fixed total area — it never fights the sum-to-one constraint.*

Proof. Immediate from $\sum_k w_{i,k} = 1$ (4) and $\sum_k r_k = (\sum_k T_k - N\bar{A})/\bar{A} = (A - A)/\bar{A} = 0$. □

Proposition 6.2 (2. Self-deactivation — no penalty ceiling)**.** *At any balanced configuration ($T_k = \bar{A}$ for all k): $r = 0$, hence $P_{\rm bal} = 0$ and, by (10), $\nabla_U P_{\rm bal} = 0$ identically. Moreover the Hessian there reduces to the positive-semidefinite Gauss–Newton block $\frac{\gamma}{\bar{A}^2} J^\top J$ (with $J_{l,(i,k)} = \partial T_l/\partial u_{i,k}$), since the second term $\frac{\gamma}{\bar{A}} \sum_l r_l \nabla^2 T_l$ vanishes with r.*

Proof. Every entry of (10) carries the factor $r_k - m_i$, which is a linear combination of the r_l; $r = 0$ annihilates it. Differentiating $\nabla P_{\rm bal} = \frac{\gamma}{\bar{A}^2} J^\top (T - \bar{A}\mathbf{1})$ gives $\nabla^2 P_{\rm bal} = \frac{\gamma}{\bar{A}^2} J^\top J + \frac{\gamma}{\bar{A}} \sum_l r_l \nabla^2 T_l$. □

Remark 6.1 (Contrast with the λ crispness penalty)**.** The crispness penalty's gradient $-\lambda \nabla \operatorname{Var}(u_k)/T_{\rm eff}$ (document 06, eq. (24)) is nonzero at every non-constant u_k — it is *constant pressure* that never switches off, which is the mechanism of the empirically observed λ ceiling (raise λ too far and the pressure dominates the perimeter terms; see `docs/reference/winner_take_all_partition_gap.md` §9). The balance force instead fades to zero *at* the balanced solutions it targets, and near them contributes only benign $\succeq 0$ curvature along the N territory directions — it cannot dominate the energy at the solution, so it has no analogous ceiling.

Proposition 6.3 (3. Interface-band localization)**.** *Let vertex i be δ-crisp for cell k: $u_{i,k} = 1 - \delta$ and $\sum_{l \neq k} u_{i,l} = \delta$, $\delta \in [0, \frac{1}{2})$. Then, with $p = 2$,*

$$1 - w_{i,k} \;\le\; \frac{\delta^2}{(1-\delta)^2}, \qquad \left|\frac{\partial P_{\rm bal}}{\partial u_{i,k}}\right| \;\le\; \frac{2\gamma v_i}{\bar{A}} \frac{\operatorname{osc}(r)\,\delta^2}{(1-\delta)^3}, \qquad \left|\frac{\partial P_{\rm bal}}{\partial u_{i,l}}\right| \;\le\; \frac{2\gamma v_i}{\bar{A}} \frac{\operatorname{osc}(r)\,\delta}{(1-\delta)^2} \quad (l \neq k).$$

In particular at an exactly crisp vertex ($\delta = 0$) the force vanishes identically in every component. The balance force is supported on the blurry interface band where $w_{i,\cdot}$ is spread — it moves fences, not interiors*.*

Proof. $1 - w_{i,k} = \sum_{l \neq k} u_{i,l}^2 / S_i \le (\sum_{l \neq k} u_{i,l})^2 / S_i \le \delta^2/(1-\delta)^2$ using $S_i \ge u_{i,k}^2 = (1-\delta)^2$. For the winner entry, in (11): $u_{i,k}/S_i \le 1/u_{i,k} = 1/(1-\delta)$ and $|r_k - m_i| = |\sum_{l \neq k} w_{i,l}(r_k - r_l)| \le \operatorname{osc}(r)\,(1 - w_{i,k}) \le \operatorname{osc}(r)\,\delta^2/(1-\delta)^2$. For a loser entry $l \neq k$: $u_{i,l} \le \delta$, so $u_{i,l}/S_i \le \delta/(1-\delta)^2$ and $|r_l - m_i| \le \operatorname{osc}(r)$. □

Proposition 6.4 (4. Neutrality at the symmetric state)**.** *At the uniform (symmetric) configuration $u_{i,k} \equiv 1/N$: $w_{i,k} = 1/N$, $T_k = \bar{A}$, $r = 0$, hence $P_{\rm bal} = 0$ and $\nabla P_{\rm bal} = 0$. The symmetric state remains a critical point of the total energy with unchanged value. The added curvature there is the Gauss–Newton block of Proposition 6.2, which vanishes on every* territory-neutral *perturbation ($Je = 0$); in particular symmetry-breaking directions that keep the soft territories equal (the ones a seeded, balanced initialization excites) feel no new resistance, and the term cannot deepen the random-init symmetric trap (`docs/reference/winner_take_all_partition_gap.md`).*

Proof. Uniform rows give $S_i = N^{1-p}$ and $w_{i,k} = N^{-p}/N^{1-p} = 1/N$, so $T_k = A/N = \bar{A}$ and $r = 0$; apply Proposition 6.2. □

Proposition 6.5 (5. Restoration of a starving cell)**.** *Let cell k be in deficit, $r_k < 0$. At every vertex i where k has partial presence ($u_{i,k} > 0$) and at least one co-present cell is less in deficit ($\exists\, l :\ w_{i,l} > 0,\ r_l > r_k$), we have $m_i > r_k$ and hence*

$$\frac{\partial P_{\rm bal}}{\partial u_{i,k}} \;<\; 0,$$

i.e. the descent direction $-\nabla$ strictly raises *$u_{i,k}$, with magnitude proportional to the local deficit contrast $m_i - r_k$. For a starving cell $T_k \to 0$ gives $r_k \to -1 \le r_l$ for all l, so the lift is active on the cell's entire halo. This is the built-in "reward for winning territory" that no term of the base energy provides.*

Proof. $m_i = \sum_l w_{i,l} r_l$ is a convex combination with weight $w_{i,k} < 1$ (since some other cell is present), so $m_i \ge r_k + w_{i,l}(r_l - r_k) > r_k$. The sign follows from (10), whose prefactor is positive when $u_{i,k} > 0$. □

6.1 6. Γ-consistency

Two statements make the consistency claim precise: the Γ-limit of the total energy is unchanged on the constrained domain (Proposition 6.6), and the soft–hard territory gap is quantitatively band-confined (Proposition 6.7).

For the continuum statement, define for measurable $u : S \to \Delta^{N-1}$ with $\int_S u_k \, dA = \bar{A}$ the continuum soft territory and balance term

$$T_k(u) = \int_S \frac{u_k^p}{\sum_l u_l^p} \, dA, \qquad P_{\mathrm{bal}}(u) = \frac{\gamma}{2} \sum_k \Big(\frac{T_k(u) - \bar{A}}{\bar{A}} \Big)^2.$$

Proposition 6.6 (Γ-limit invariance). *Let $E_\varepsilon(u) = \sum_k F_\varepsilon(u_k)$ be the Modica–Mortola partition energy, which Γ-converges in $L^1(S)^N$ as $\varepsilon \to 0$ to $F_0(u) = 2c_W \cdot \frac{1}{2} \operatorname{Per}(u)$ on partition indicators $u = \chi = (\chi_{\Omega_1}, \ldots, \chi_{\Omega_N})$ and $+\infty$ otherwise [4, 3, 1]. Then:*

1. *P_{bal} is continuous on the admissible set with respect to L^1 convergence;*
2. *hence $E_\varepsilon + P_{\mathrm{bal}}$ Γ-converges to $F_0 + P_{\mathrm{bal}}$ [2, Remark 1.7];*
3. *on the limit domain the equal-area constraint forces $|\Omega_k| = \int_S \chi_k \, dA = \bar{A}$, and for indicators $T_k(\chi) = |\Omega_k|$, so $P_{\mathrm{bal}}(\chi) = 0$ there:* the constrained Γ-limit is the unchanged perimeter functional.

Proof. (1) The map $g : \Delta^{N-1} \to \Delta^{N-1}$, $g_k(z) = z_k^p / \sum_l z_l^p$, is continuous and bounded on the compact simplex (the denominator is $\geq N^{1-p}$ by Proposition 3.1). If $u^n \to u$ in L^1, every subsequence has a further subsequence converging a.e.; along it $g(u^n) \to g(u)$ a.e., and $|g| \leq 1$ allows dominated convergence, so $T_k(u^n) \to T_k(u)$ along that sub-subsequence. Since the limit is independent of the subsequence, the full sequence converges, and P_{bal}, a polynomial in the T_k, converges too. (2) is the standard stability of Γ-limits under continuous additive perturbations. (3) For an indicator row $\chi_{i,\cdot} \in \{e_1, \ldots, e_N\}$, $g(\chi) = \chi$ exactly, so $T_k(\chi) = \int \chi_k = |\Omega_k|$; the constraint $\int \chi_k = \bar{A}$ gives $r = 0$. □

Proposition 6.7 (Quantitative soft–hard gap along recovery fronts)**.** *Let u^ε be the standard Modica–Mortola recovery configuration for a two-cell interface of length* Per_k*: the logistic profile $\phi(t) = (1 + e^{-t/\varepsilon})^{-1}$ across the front (the heteroclinic of the quartic well $W(s) = s^2(1-s)^2$). Then for $p = 2$*

$$\big|T_k(u^\varepsilon) - T_k^{\mathrm{wta}}(u^\varepsilon)\big| \;\leq\; C_2 \, \varepsilon \operatorname{Per}_k + o(\varepsilon), \qquad C_2 = \int_{-\infty}^{\infty} \big|g_k(\phi(s)) - H(s)\big| \, ds = \ln 2, \tag{13}$$

with H the Heaviside step. Moreover, the signed *leading-order gap across a straight symmetric front vanishes: by the symmetry $g_k(1-z) = 1 - g_k(z)$ of the two-cell weight $g_k(z) = z^2/(z^2 + (1-z)^2)$, the surplus on the winning side cancels the deficit on the losing side exactly, so the $O(\varepsilon)$ bound (13) is attained only through curvature and junction effects.*

Proof. Write the gap as an integral over tubular coordinates around the interface; transversally, with s the scaled signed distance, $T_k - T_k^{\mathrm{wta}} = \varepsilon \operatorname{Per}_k \int_{\mathbb{R}} \big(g_k(\phi(s)) - H(s)\big) \, ds + o(\varepsilon)$ (coarea; the profile relaxes to 0/1 exponentially so the integral converges absolutely: for $s > 0$, $1 - g_k(\phi) = (1-\phi)^2/(\phi^2 + (1-\phi)^2) \leq ((1-\phi)/\phi)^2 = e^{-2s}$). For the constant: substitute $t = e^{-s}$, $1 - g_k = t^2/(1+t^2)$, $ds = dt/t$, so $\int_0^\infty (1 - g_k(\phi(s))) \, ds = \int_0^1 \frac{t}{1+t^2} \, dt = \frac{1}{2} \ln 2$; doubling for the two sides gives $C_2 = \ln 2$. The signed cancellation follows from $g_k(1-z) = (1-z)^2/((1-z)^2 + z^2) = 1 - g_k(z)$. □

Together: adding P_{bal} leaves the $\varepsilon \to 0$ perimeter Γ-limit untouched on the equal-area constraint manifold, and at finite ε the surrogate T_k tracks the delivered WTA territory to within an $O(\varepsilon \operatorname{Per}_k)$ band-confined error — the same order as the interface width itself.

7 Calibration of the strength γ

Scaling. By (11) a band-vertex entry of the balance gradient has magnitude $\sim \gamma\, v_i\,(u/S)/\bar{A}$. With $v_i \sim A/|V|$ and $\bar{A} = A/N$, this is $\sim \gamma\, N/|V| = \gamma/$(vertices per cell): under the mesh-budget policy that holds vertices-per-cell fixed as N grows (`docs/plans/PHASE1_N1000_VALIDITY_PLAN.md` P3), the force is *N-invariant* and one calibrated γ transfers across N. γ is calibrated **once**, against the double-well force it must perturb but not overwhelm, and is *not* retuned per N.

Procedure (recorded; script `scripts/debug_archive/calibrate_wta_gamma.py`). On a converged production solution, collect the interface-band vertices (winning density $u_{\mathrm{win}} < 0.9$; the force is supported there by Proposition 6.3). At each band vertex compute (i) the local double-well gradient magnitude $\left|\frac{2}{\varepsilon}(Mq_k)_i(1-2u_{i,k})\right|$ of the winning cell, and (ii) the balance-force magnitude that a synthetic 5% territory deficit of the winning cell would produce there, $\frac{2\gamma}{\bar{A}} v_i \frac{u_{i,k}}{S_i} \cdot 0.05\,(1-w_{i,k})$ (all other cells balanced). Choose γ so the *median* ratio (ii)/(i) over the band equals 10% — the middle of the plan's 5–15% target window: strong enough to move fences on a 5% deficit, weak enough never to compete with the well away from balance.

Measured calibration (N=300 reference run `run_20260714_224821`, $|V| = 47{,}488$, $\varepsilon = 0.0158$, $\bar{A} = 7.895 \times 10^{-2}$). The band ($u_{\mathrm{win}} < 0.9$) holds 18,974 vertices (40.0% of $|V|$). The per-unit-γ force ratio on the band has median 1.453×10^{-2} (interquartile range $[4.17 \times 10^{-3}, 6.85 \times 10^{-2}]$), giving $\gamma = 0.10/1.453 \times 10^{-2} = 6.88$ for an exact 10% median. The calibrated default is $\boldsymbol{\gamma = 7.0}$ (median ratio 10.2%; config `wta_balance_gamma`); the run configs under `parameters/` carry it explicitly. At fixed mesh the band-force ratio scales like $\gamma\varepsilon N$ (the well force per vertex is $\sim v_i/\varepsilon$), so on the *same* finest mesh the Stage-6 $N = 200$ configuration sees a median ratio of $\approx 6.8\%$ (a factor 200/300 smaller) — still inside the 5–15% window.

8 The discrete-area trim

The balance term controls the *soft* territory; the residual soft–hard gap (Proposition 6.7) is closed by a derivative-free controller on the projection targets. The projection (`src/optimization/projection.py`) already accepts an arbitrary feasible target vector d with $\sum_k d_k = A$; the trim retargets it every J accepted PGD iterations (default $J = 200$, `wta_trim_period`):

$$\tilde{d}_k \;=\; \mathrm{clip}\Big(d_k + \beta\,(\bar{A} - T_k^{\mathrm{wta}}),\ \bar{A}(1-c),\ \bar{A}(1+c)\Big), \qquad d \;\leftarrow\; \tilde{d} \cdot \frac{A}{\sum_k \tilde{d}_k}, \tag{14}$$

with damping $\beta = 0.5$ (`wta_trim_damping`) and clamp $c = 0.20$ (`wta_trim_clamp`). T^{wta} is measured by the existing argmax machinery (`detect_area_imbalance`, `src/partition/find_contours.py`); the update costs $O(|V|)$ per application, amortized to nothing.

Proposition 8.1 (Fixed point of the trim). *Suppose $\sum_k d_k = A$ before an update. If the clamp in (14) is inactive, then $\sum_k \tilde{d}_k = A$ (because $\sum_k T_k^{\mathrm{wta}} = A$ identically), so the renormalization is the identity, and d is a fixed point of the update iff $T_k^{\mathrm{wta}} = \bar{A}$ for every k —* exact discrete equality*, independent of any residual soft–hard surrogate gap.*

Proof. $\sum_k \tilde{d}_k = \sum_k d_k + \beta(N\bar{A} - \sum_k T_k^{\mathrm{wta}}) = A + \beta(A - A) = A$. With the renormalization the identity, $\tilde{d} = d$ iff $\beta(\bar{A} - T_k^{\mathrm{wta}}) = 0$ for all k. ☐

Remark 8.1 (What the trim trades away). At a fixed point the *continuous* masses equal $d \neq \bar{A}\mathbf{1}$ in general — they are deliberately unpinned by up to the clamp $c\,\bar{A}$, in exchange for exact *discrete* equality. Phase 2 is indifferent: its iteration-0 equal-area feasibility is exactly the discrete equality the trim delivers (the worst-cell discrete deviation *is* the Phase 2 initial constraint violation; see `docs/reference/winner_take_all_partition_gap.md` §7). The damped Picard update has no global convergence proof on the nonconvex inner problem; the clamp and damping keep it a bounded perturbation, and the per-level reset below keeps levels independent.

Lifecycle. d is carried *within* a refinement level and reset to $\bar{A}\mathbf{1}$ at the start of each new level (after interpolation): the mesh, and with it T^{wta}, changes across levels. In the implementation this reset is structural — each level constructs a fresh optimizer whose `optimize` call re-initializes d.

Code: discrete-area trim

`src/optimization/pgd_optimizer.py`: `_apply_wta_trim` (called from `optimize` every `wta_trim_period` accepted iterations, guarded by `wta_trim_enabled`; default off)

```
T_wta = np.asarray(detect_area_imbalance(A, v, N)['discrete_areas'])
d = np.clip(d + beta * (Abar - T_wta), Abar*(1-c), Abar*(1+c))
d *= total_area / d.sum()
```

Config: `wta_trim_enabled`, `wta_trim_period`, `wta_trim_damping`, `wta_trim_clamp` in `RelaxationConfig`.

Summary table

Quantity	Form	Method	Implementing function
Soft weights (4)	$w_{i,k} = u_{i,k}^p / S_i$	analytical	`compute_energy`
Soft territory (5)	$T_k = \sum_i v_i w_{i,k}$	analytical	`compute_energy`
Balance penalty (7)	$\frac{\gamma}{2}\sum_k r_k^2$, $r_k = (T_k - \bar{A})/\bar{A}$	analytical	`compute_energy`
its gradient (10)	$\frac{\gamma p}{\bar{A}} v_i \frac{u_{i,k}^{p-1}}{S_i}(r_k - m_i)$	analytical	`compute_gradient`
Normalizer bounds (6)	$N^{1-p} \le S_i \le 1$	analytical	(stability guard)
Gradient bound (12)	$\le \frac{2\gamma v_i}{\bar{A}}\sqrt{N}\,\mathrm{osc}(r)$	analytical	— (theory)
Soft–hard gap (13)	$\le \varepsilon \ln 2\, \mathrm{Per}_k + o(\varepsilon)$	analytical	— (theory)
Discrete trim (14)	damped clip'd retarget of d	controller	`_apply_wta_trim`
FD verification	central difference, rel. err. $< 10^{-6}$	numerical	`testing/test_wta_balance_...`

The balance gradient (10) is analytical and verified against Richardson-extrapolated central finite differences on deterministic feasible densities (`testing/test_wta_balance_gradient_analytical.py`); the base energy E_0 and its gradient are unchanged (`docs/math/06-phase1-energy-discretization/`).

References

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